B1 Fluids, 2025 — Model Solutions

Question 1 — Streamfunction, vorticity and point vortices

Problem statement. Introduce the streamfunction ψ for 2D incompressible flow and prove the Bernoulli–vorticity relations $\omega = \omega(\psi)$, $B = B(\psi)$, $\omega = -\mathrm{d}B/\mathrm{d}\psi$. Apply to a Rankine vortex, and to two point vortices of strengths A and B.

(a) Continuity, streamfunction and vorticity

Continuity for a compressible fluid:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0. \tag{1}$$

Incompressibility ($\mathrm{D}\rho/\mathrm{D}t = 0$) reduces this to

$$\nabla \cdot \mathbf{v} = 0.$$

In 2D, $\partial_x v_x + \partial_y v_y = 0$ is the integrability condition for $\psi(x, y)$ with

$$v_x = \frac{\partial \psi}{\partial y}, \qquad v_y = -\frac{\partial \psi}{\partial x},$$

i.e. $\mathbf{v} = -\hat{\mathbf{z}} \times \nabla \psi$. Compute the vorticity $\boldsymbol{\omega} = \omega \hat{\mathbf{z}}$:

$$\omega = \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} = -\frac{\partial^2 \psi}{\partial x^2} - \frac{\partial^2 \psi}{\partial y^2}.$$

Hence

$$\boxed{\omega = -\nabla^2 \psi.} \tag{2}$$

Lines of constant ψ are streamlines since $\mathbf{v} \cdot \nabla \psi = (\partial_y \psi)(\partial_x \psi) + (-\partial_x \psi)(\partial_y \psi) = 0$.

(b) Bernoulli–vorticity theorem for steady 2D inviscid flow

Euler (gravity absorbed into p):

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} = -\frac{1}{\rho} \nabla p. \tag{3}$$

Apply $(\mathbf{v} \cdot \nabla)\mathbf{v} = \nabla(\frac{1}{2} v^2) - \mathbf{v} \times (\nabla \times \mathbf{v})$ for steady flow:

$$\nabla \left(\frac{p}{\rho} + \frac{v^2}{2} \right) = \mathbf{v} \times (\omega \hat{\mathbf{z}}).$$

Compute $\mathbf{v} \times \hat{\mathbf{z}}$ in components:

$$\mathbf{v} \times \hat{\mathbf{z}} = (v_y, -v_x, 0) = (-\partial_x \psi, -\partial_y \psi, 0) = -\nabla \psi.$$

Hence with $B \equiv p/\rho + \frac{1}{2}v^2$,

$$\nabla B = -\omega \nabla \psi. \tag{4}$$

Dot with $\mathbf{v}$ (using $\mathbf{v} \cdot \nabla \psi = 0$):

$$\mathbf{v} \cdot \nabla B = 0.$$

So B is constant along streamlines:

$$\boxed{B = B(\psi).} \tag{5}$$

Take the curl of (3); both gradient terms vanish, leaving $\mathrm{D}\omega/\mathrm{D}t = 0$. In steady flow $\mathbf{v} \cdot \nabla \omega = 0$, so ω is constant along streamlines:

$$\boxed{\omega = \omega(\psi).} \tag{6}$$

Substitute $\nabla B = (\mathrm{d}B/\mathrm{d}\psi)\nabla\psi$ into (4):

$$\frac{\mathrm{d}B}{\mathrm{d}\psi}\nabla\psi = -\omega\nabla\psi.$$

Cancel $\nabla\psi$:

$$\boxed{\omega = -\frac{\mathrm{d}B}{\mathrm{d}\psi}.} \tag{7}$$

(c) Rankine vortex

Symmetry gives $\mathbf{v} = v_\theta(r)\hat{\boldsymbol{\theta}}$ and

$$\omega = \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}(r v_\theta).$$

Inside ($r \le r_0$, $\omega = \omega_0$), regularity at $r = 0$:

$$v_\theta = \tfrac{1}{2}\omega_0 r.$$

Outside ($\omega = 0$), $r v_\theta =$ const; matching at r_0:

$$\boxed{v_\theta(r) = \begin{cases} \frac{1}{2}\omega_0 r, & r \le r_0, \\ \frac{1}{2}\omega_0 r_0^2/r, & r > r_0. \end{cases}} \tag{8}$$

Radial Euler (centripetal balance):

$$\frac{1}{\rho}\frac{\mathrm{d}p}{\mathrm{d}r} = \frac{v_\theta^2}{r}.$$

Outside, substitute $v_\theta = \omega_0 r_0^2/(2r)$:

$$\frac{v_\theta^2}{r} = \frac{\omega_0^2 r_0^4}{4r^3}.$$

Integrate from r to ∞ with $p(\infty) = p_\infty$:

$$p_\infty - p(r) = \int_r^\infty \frac{\rho\omega_0^2 r_0^4}{4r'^3}\,\mathrm{d}r'.$$

Evaluate the integral:

$$p_\infty - p(r) = \frac{\rho\omega_0^2 r_0^4}{4}\left[-\frac{1}{2r'^2}\right]_r^\infty = \frac{\rho\omega_0^2 r_0^4}{8r^2}.$$

Hence outside,

$$p(r) = p_\infty - \frac{\rho\omega_0^2 r_0^4}{8r^2}.$$

Inside, substitute $v_\theta = \frac{1}{2}\omega_0 r$:

$$\frac{v_\theta^2}{r} = \tfrac{1}{4}\omega_0^2 r.$$

Integrate $\partial_r p = \frac{1}{4}\rho\omega_0^2 r$:

$$p(r) = \tfrac{1}{8}\rho\omega_0^2 r^2 + C.$$

Match p across $r = r_0$:

$$\tfrac{1}{8}\rho\omega_0^2 r_0^2 + C = p_\infty - \tfrac{1}{8}\rho\omega_0^2 r_0^2.$$

Solve for C:

$$C = p_\infty - \tfrac{1}{4}\rho\omega_0^2 r_0^2.$$

Therefore

$$\boxed{p(r) = \begin{cases} p_\infty - \frac{1}{4}\rho\omega_0^2 r_0^2 + \frac{1}{8}\rho\omega_0^2 r^2, & r \le r_0, \\ p_\infty - \dfrac{\rho\omega_0^2 r_0^4}{8r^2}, & r > r_0. \end{cases}} \tag{9}$$

(d) Two point vortices

Each vortex moves only in the field of the other. A point vortex of strength Γ induces $v_\theta = \Gamma/(2\pi r)$ at distance r. With separation $\mathbf{d} = \mathbf{x}_2 - \mathbf{x}_1$, $L = |\mathbf{d}|$:

$$\dot{\mathbf{x}}_1 = -\frac{B}{2\pi L^2}\hat{\mathbf{z}} \times \mathbf{d}, \qquad \dot{\mathbf{x}}_2 = \frac{A}{2\pi L^2}\hat{\mathbf{z}} \times \mathbf{d}. \tag{10}$$

Subtract to get the separation evolution:

$$\dot{\mathbf{d}} = \frac{A+B}{2\pi L^2}\hat{\mathbf{z}} \times \mathbf{d}. \tag{11}$$

Form the weighted centroid:

$$A\dot{\mathbf{x}}_1 + B\dot{\mathbf{x}}_2 = \frac{-AB + AB}{2\pi L^2}\hat{\mathbf{z}} \times \mathbf{d} = 0, \qquad \text{so} \quad A\mathbf{x}_1 + B\mathbf{x}_2 = \text{const.} \tag{12}$$

Since $\dot{\mathbf{d}} \perp \mathbf{d}$, $|\mathbf{d}| = L$ is constant.

Case $A = B$. Centroid fixed; both vortices co-rotate about the midpoint on circles of radius $L/2$.

Case $A = -B$. From (11), $\dot{\mathbf{d}} = 0$. Substitute $A = -B$:

$$\dot{\mathbf{x}}_1 = -\frac{B}{2\pi L^2}\hat{\mathbf{z}} \times \mathbf{d}, \qquad \dot{\mathbf{x}}_2 = -\frac{B}{2\pi L^2}\hat{\mathbf{z}} \times \mathbf{d}.$$

Both vortices have equal velocity, so the pair translates along parallel straight lines perpendicular to $\mathbf{d}$.

Sketch.

$A = B > 0$ (co-rotating CCW) $\qquad\qquad$ $A = -B$ (translating)

Frequency (periodic case $A = B$). From (11), $\mathbf{d}$ rotates uniformly with angular speed

$$\Omega = \frac{A+B}{2\pi L^2}.$$

Set $A = B$:

$$\boxed{\omega_{\text{orb}} = \frac{A+B}{2\pi L^2} = \frac{A}{\pi L^2} \quad (A = B).} \tag{13}$$

Question 2 — Complex potential and conformal mappings

Problem statement. For analytic $w(z) = \phi + \mathrm{i}\psi$, derive Cauchy–Riemann, harmonicity and orthogonality of level curves. Apply to $w = Bz^2$. Define a conformal mapping; invert $Z = z/2 + \sqrt{z^2/4 - c^2}$ and find Φ, Ψ for $w(z) = Az$ in the Z-plane.

(a) Analytic functions and Cauchy–Riemann

Analytic w: $w'(z)$ exists independently of the direction of $\Delta z \to 0$. Take $\Delta z = \Delta x$:

$$w'(z) = \frac{\partial \phi}{\partial x} + \mathrm{i}\frac{\partial \psi}{\partial x}.$$

Take $\Delta z = \mathrm{i}\Delta y$:

$$w'(z) = -\mathrm{i}\frac{\partial \phi}{\partial y} + \frac{\partial \psi}{\partial y}.$$

Equate real and imaginary parts:

$$\boxed{\frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y}, \qquad \frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x}.} \tag{14}$$

Differentiate the first of (14) in x and the second in y:

$$\frac{\partial^2 \phi}{\partial x^2} = \frac{\partial}{\partial x}\frac{\partial \psi}{\partial y}, \qquad \frac{\partial^2 \phi}{\partial y^2} = -\frac{\partial}{\partial y}\frac{\partial \psi}{\partial x}.$$

Add the two:

$$\nabla^2 \phi = 0.$$

Symmetrically, $\nabla^2 \psi = 0$. Compute the dot product using (14):

$$\nabla\phi \cdot \nabla\psi = (\partial_x \phi)(\partial_x \psi) + (\partial_y \phi)(\partial_y \psi) = (\partial_y \psi)(\partial_x \psi) + (-\partial_x \psi)(\partial_y \psi) = 0.$$

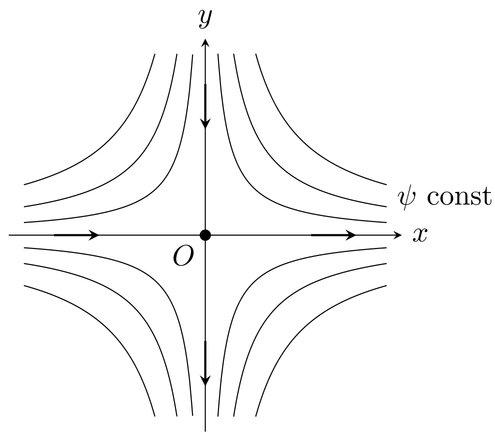
For 2D incompressible irrotational inviscid flow, ϕ is the velocity potential ($\mathbf{v} = \nabla\phi$) and ψ the streamfunction; (14) reproduces $v_x = \partial_x \phi = \partial_y \psi$, $v_y = \partial_y \phi = -\partial_x \psi$.

(b) The flow $w(z) = Bz^2$

$w = B(x^2 - y^2) + 2\mathrm{i}Bxy$, so

$$\boxed{\phi = B(x^2 - y^2), \qquad \psi = 2Bxy.} \tag{15}$$

Streamlines $\psi = \text{const}$ are hyperbolae $xy = \text{const}$ ($\psi = 0$ is the axes); equipotentials $x^2 - y^2 = \text{const}$ are the orthogonal family.



For $B > 0$: in the first quadrant fluid descends along the $+y$ axis and exits along the $+x$ axis.

Velocity: $u - \mathrm{i}v = w'(z) = 2Bz$, so $\mathbf{v} = (2Bx, -2By)$. The point $w'(z) = 0$ at $z = 0$ is a stagnation point: 2D corner-flow saddle, fluid against a right-angled wall.

(c) Conformal mapping and Joukowski plate

A holomorphic $Z = f(z)$ with $f' \neq 0$ is a conformal map: angles preserved. If $w(z)$ is analytic then $W(Z) \equiv w(F(Z))$ with $z = F(Z)$ is analytic in Z and represents another potential flow; streamlines map by f.

Invert $Z = z/2 + \sqrt{z^2/4 - c^2}$. Subtract $z/2$:

$$Z - \tfrac{z}{2} = \sqrt{\tfrac{z^2}{4} - c^2}.$$

Square both sides:

$$\left(Z - \tfrac{z}{2}\right)^2 = \tfrac{z^2}{4} - c^2.$$

Expand the left-hand side:

$$Z^2 - zZ + \tfrac{z^2}{4} = \tfrac{z^2}{4} - c^2.$$

Cancel $z^2/4$:

$$Z^2 - zZ = -c^2.$$

Solve for z:

$$zZ = Z^2 + c^2.$$

Divide by Z:

$$\boxed{z = F(Z) = Z + \frac{c^2}{Z}.} \tag{16}$$

On $|Z| = c$, write $Z = c\mathrm{e}^{\mathrm{i}\theta}$:

$$z = c\mathrm{e}^{\mathrm{i}\theta} + \frac{c^2}{c\mathrm{e}^{\mathrm{i}\theta}} = c\mathrm{e}^{\mathrm{i}\theta} + c\mathrm{e}^{-\mathrm{i}\theta}.$$

Apply $e^{i\theta} + e^{-i\theta} = 2\cos\theta$:

$$z = 2c\cos\theta \in [-2c, 2c].$$

For $w(z) = Az$ (uniform stream of speed A), compose with $z = F(Z)$:

$$W(Z) = A\left(Z + \frac{c^2}{Z}\right).$$

Write $Z = Re^{i\Theta}$:

$$W = A\left(Re^{i\Theta} + \frac{c^2}{R}e^{-i\Theta}\right).$$

Expand using Euler:

$$W = A\left(R + \frac{c^2}{R}\right)\cos\Theta + iA\left(R - \frac{c^2}{R}\right)\sin\Theta.$$

Real and imaginary parts:

$$\boxed{\Phi = A\left(R + \frac{c^2}{R}\right)\cos\Theta, \qquad \Psi = A\left(R - \frac{c^2}{R}\right)\sin\Theta.} \tag{17}$$

Identify the $\Psi = 0$ streamlines: the real axis $\Theta = 0, \pi$ *and* the circle $R = c$. Hence (17) represents uniform flow of speed A past a rigid cylinder of radius c in the Z-plane. Under $z = Z + c^2/Z$ the cylinder $|Z| = c$ maps to the segment $z \in [-2c, 2c]$, so the same physical flow appears in the z-plane as $w = Az$, i.e. uniform flow past a horizontal flat plate of length $4c$.

Real fluid. Viscosity enforces no-slip, generating a boundary layer that separates on the rear half at large Re, producing a wake and finite drag (resolving d'Alembert's paradox). The wake is unsteady (von Kármán street), so the symmetric streamlines of (17) are never observed near the body; the conformal result describes only the outer irrotational flow.

Question 3 — Thin viscous film on an inclined conveyor belt

Problem statement. A thin viscous film $h(x,t)$ on a belt inclined at θ, moving at $U_0\hat{\mathbf{x}}$; y normal to belt. Define Re, derive the lubrication equation, find $v_x(y)$, the equilibrium thickness, and timescales after the belt stops.

(a) Reynolds number and lubrication

$$\mathrm{Re} = \frac{|(\mathbf{v}\cdot\nabla)\mathbf{v}|}{|\nu\nabla^2\mathbf{v}|} \sim \frac{UL}{\nu}.$$

$\mathrm{Re} \ll 1$: viscous diffusion dominates (Stokes). For a thin film, aspect ratio $\epsilon = h/L \ll 1$; incompressibility gives $v_y \sim \epsilon U_0$, and the inertial term is smaller than $\nu\partial_y^2 v_x$ by $\epsilon\,\mathrm{Re} \ll 1$:

$$0 = -\frac{1}{\rho}\frac{\partial p}{\partial x} + g\sin\theta + \nu\frac{\partial^2 v_x}{\partial y^2}, \qquad 0 = -\frac{1}{\rho}\frac{\partial p}{\partial y} - g\cos\theta. \tag{18}$$

Streamwise viscous term $\nu\partial_x^2 v_x$ suppressed by ϵ^2.

Integrate $(18)_2$ in y:

$$p(x,y) = -\rho g\cos\theta\, y + f(x).$$

Apply $p(y=h)=p_a$ (no surface tension):

$$f(x)=p_a+\rho g\cos\theta\, h(x).$$

Hence

$$p=p_a+\rho g\cos\theta\,(h-y).$$

Differentiate with respect to x:

$$\frac{\partial p}{\partial x}=\rho g\cos\theta\,\frac{\partial h}{\partial x}.$$

Substitute into $(18)_1$:

$$\boxed{g\frac{\partial h}{\partial x}\cos\theta+g\sin\theta=\nu\frac{\partial^2 v_x}{\partial y^2}.} \tag{19}$$

(b) Boundary conditions and velocity profile

Two boundary conditions: no-slip on the belt (v_x matches the belt speed), and zero tangential viscous stress at the free surface (the air above exerts negligible drag):

$$v_x(0)=U_0,\qquad \left.\frac{\partial v_x}{\partial y}\right|_{y=h}=0.$$

Write $G\equiv g\cos\theta\,\partial h/\partial x+g\sin\theta$, so (19) reads $\nu\,\partial^2 v_x/\partial y^2=G$. Integrate once in y:

$$\nu\frac{\partial v_x}{\partial y}=Gy+C_1.$$

Apply $\partial_y v_x|_{y=h}=0$:

$$C_1=-Gh.$$

So

$$\nu\frac{\partial v_x}{\partial y}=G(y-h).$$

Integrate again:

$$\nu v_x=G\big(\tfrac{1}{2}y^2-hy\big)+C_2.$$

Apply $v_x(0)=U_0$:

$$C_2=\nu U_0.$$

Therefore

$$\boxed{v_x(x,y,t)=U_0+\frac{1}{\nu}\left[g\cos\theta\,\frac{\partial h}{\partial x}+g\sin\theta\right]\big(\tfrac{1}{2}y^2-hy\big)\,.} \tag{20}$$

(c) Equilibrium thickness

Steady, $\partial h/\partial x=0$:

$$v_x(y)=U_0+\frac{g\sin\theta}{\nu}\big(\tfrac{1}{2}y^2-h_{\mathrm{eq}}y\big)\,. \tag{21}$$

Flux per unit width:

$$Q=\int_0^{h_{\mathrm{eq}}}v_x\,\mathrm{d}y.$$

Substitute (21):

$$Q = \int_0^{h_{\rm eq}} \left[U_0 + \frac{g\sin\theta}{\nu}\left(\tfrac{1}{2}y^2 - h_{\rm eq}y\right)\right] \mathrm{d}y.$$

Integrate term by term:

$$Q = U_0 h_{\rm eq} + \frac{g\sin\theta}{\nu}\left(\tfrac{1}{6}h_{\rm eq}^3 - \tfrac{1}{2}h_{\rm eq}^3\right).$$

Simplify:

$$Q = U_0 h_{\rm eq} - \frac{g\sin\theta\, h_{\rm eq}^3}{3\nu}.$$

Impose no net transport, $Q = 0$:

$$U_0 = \frac{g\sin\theta\, h_{\rm eq}^2}{3\nu}.$$

Solve for $h_{\rm eq}$:

$$\boxed{h_{\rm eq} = \sqrt{\frac{3\nu U_0}{g\sin\theta}}.} \tag{22}$$

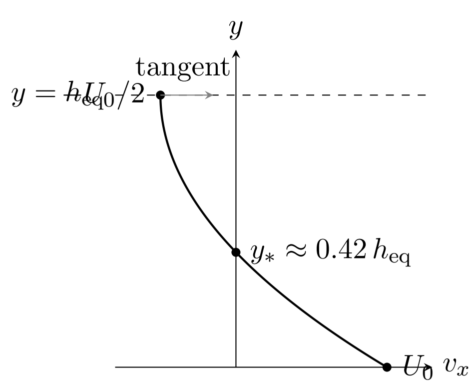
At $y = h_{\rm eq}$, evaluate v_x from (21):

$$v_x(h_{\rm eq}) = U_0 + \frac{g\sin\theta}{\nu}\left(\tfrac{1}{2}h_{\rm eq}^2 - h_{\rm eq}^2\right) = U_0 - \frac{g\sin\theta\, h_{\rm eq}^2}{2\nu}.$$

Use $g\sin\theta\, h_{\rm eq}^2/\nu = 3U_0$:

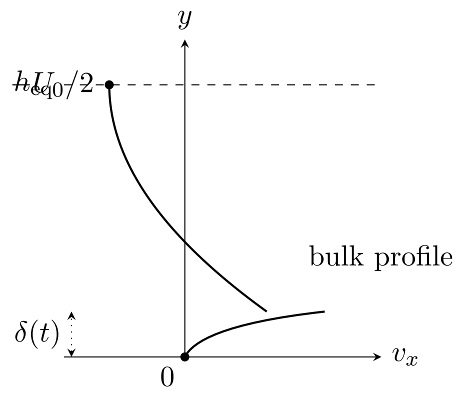
$$v_x(h_{\rm eq}) = U_0 - \tfrac{3}{2}U_0 = -\tfrac{1}{2}U_0.$$

Solve $v_x(y_*) = 0$ to find the zero crossing $y_* = h_{\rm eq}(1 - 1/\sqrt{3}) \approx 0.42\, h_{\rm eq}$.



(d) Belt switched off

Just after stop, no-slip gives $v_x(0) = 0$; momentum redistributes via a thin viscous layer $\delta(t) \sim \sqrt{\nu t}$ at the wall while the bulk retains (21).



Sketch (i): equilibrium parabola with a thin viscous kink near $y = 0$ pulling v_x to 0.

Once $\delta \sim h_{\text{eq}}$, the wall layer fills the film. Integrate $\nu \partial_y^2 v_x = -g \sin\theta$ once:

$$\nu \frac{\partial v_x}{\partial y} = -g \sin\theta \, y + C_1.$$

Apply $\partial_y v_x(h) = 0$:

$$C_1 = g \sin\theta \, h.$$

Integrate again:

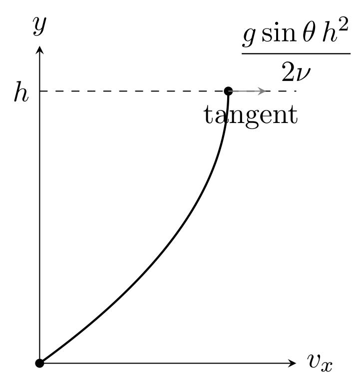
$$\nu v_x = g \sin\theta \big(hy - \tfrac{1}{2} y^2\big) + C_2.$$

Apply $v_x(0) = 0$:

$$C_2 = 0.$$

Hence

$$v_x(y) = \frac{g \sin\theta}{\nu} \left(hy - \tfrac{1}{2} y^2\right). \tag{23}$$



Sketch (ii): drainage parabola from 0 at $y = 0$ to maximum $g \sin\theta \, h^2/(2\nu)$ at $y = h$, with horizontal tangent at the free surface.

Diffusion timescale. $\partial_t v_x = \nu \partial_y^2 v_x$ across h_{eq}:

$$\boxed{\tau_{\text{diff}} \sim \frac{h_{\text{eq}}^2}{\nu}.} \tag{24}$$

Drainage timescale. Surface speed $U_{\text{drain}} \sim g \sin\theta\, h_{\text{eq}}^2/\nu$; using (22), $g \sin\theta\, h_{\text{eq}}^2 = 3\nu U_0$:

$$\boxed{\tau_{\text{drain}} \sim \frac{L\nu}{g \sin\theta\, h_{\text{eq}}^2} \sim \frac{L}{U_0}.} \tag{25}$$

$\tau_{\text{drain}}/\tau_{\text{diff}} \sim L/h_{\text{eq}} \gg 1$ (lubrication limit): fast diffusion sets up the drainage profile, slow drainage empties the film.

Question 4 — Stratified instability and Rayleigh–Bénard convection

Problem statement. (a) Derive the buoyancy frequency for incompressible $\rho(z)$ and discuss instability. (b) Derive the perturbation vorticity equation and the sixth-order equation for v_z' in Boussinesq Rayleigh–Bénard. (c) For the mode $v_z' = A\sin(kx)\sin(\pi z/H)\mathrm{e}^{\sigma t}$, find σ and the marginal Rayleigh number.

(a) Brunt–Väisälä frequency

A parcel displaced by ζ keeps density $\rho(z_0)$; surroundings have

$$\rho(z_0 + \zeta) \approx \rho(z_0) + \zeta\,\frac{\mathrm{d}\rho}{\mathrm{d}z}.$$

Buoyancy force per unit volume (upward = displaced minus parcel times g):

$$F_b = \big(\rho_{\text{surr}} - \rho_{\text{parcel}}\big)g = g\zeta\,\frac{\mathrm{d}\rho}{\mathrm{d}z}.$$

Apply Newton's second law $\rho\ddot{\zeta} = F_b$:

$$\ddot{\zeta} = \frac{g}{\rho}\frac{\mathrm{d}\rho}{\mathrm{d}z}\zeta.$$

Identify with $\ddot{\zeta} = -\omega^2\zeta$:

$$\boxed{\omega^2 = -\frac{g}{\rho}\frac{\mathrm{d}\rho}{\mathrm{d}z}.} \tag{26}$$

Stable ($\mathrm{d}\rho/\mathrm{d}z < 0$): oscillation at this frequency. Unstable ($\mathrm{d}\rho/\mathrm{d}z > 0$): $\omega^2 < 0$, exponential growth $\zeta \propto \mathrm{e}^{|\omega|t}$ (Rayleigh–Taylor instability).

(b) Equilibrium and perturbation equations

Equilibrium: $\kappa\nabla^2 T = 0$, so

$$\boxed{T_{\text{eq}}(z) = T_0 - \frac{\Delta T}{H}z.} \tag{27}$$

With $b = g\alpha(T - T_0)$, hydrostatic balance gives $p_{\text{eq}}(z) = p_0 - \rho_0 g\alpha\Delta T\, z^2/(2H)$.

Linearising $\mathbf{v} = (v_x', v_z')$, $T = T_{\text{eq}} + T'$, $p = p_{\text{eq}} + p'$:

$$\frac{\partial v_x'}{\partial t} + \frac{1}{\rho_0}\frac{\partial p'}{\partial x} = \nu\nabla^2 v_x', \tag{28}$$

$$\frac{\partial v_z'}{\partial t} - g\alpha T' + \frac{1}{\rho_0}\frac{\partial p'}{\partial z} = \nu\nabla^2 v_z', \tag{29}$$

$$\frac{\partial T'}{\partial t} - \frac{\Delta T}{H}v_z' = \kappa\nabla^2 T', \tag{30}$$

$$\frac{\partial v_x'}{\partial x} + \frac{\partial v_z'}{\partial z} = 0. \tag{31}$$

Take $\partial_z(28)-\partial_x(29)$ to eliminate p':

$$\boxed{\left(\frac{\partial}{\partial t}-\nu\nabla^2\right)\left(\frac{\partial v_x'}{\partial z}-\frac{\partial v_z'}{\partial x}\right)=-g\alpha\frac{\partial T'}{\partial x}.} \tag{32}$$

The bracket on the left is the (negative of the) perturbation vorticity; the LHS is its viscous advection rate and the RHS is the baroclinic buoyancy torque.

Apply ∂_x to (32):

$$\left(\frac{\partial}{\partial t}-\nu\nabla^2\right)(\partial_{xz}v_x'-\partial_x^2 v_z')=-g\alpha\frac{\partial^2 T'}{\partial x^2}.$$

Use (31) as $\partial_x v_x'=-\partial_z v_z'$, so $\partial_{xz}v_x'=-\partial_z^2 v_z'$:

$$\partial_{xz}v_x'-\partial_x^2 v_z'=-\partial_z^2 v_z'-\partial_x^2 v_z'=-\nabla^2 v_z'.$$

Substitute:

$$\left(\frac{\partial}{\partial t}-\nu\nabla^2\right)\nabla^2 v_z'=g\alpha\frac{\partial^2 T'}{\partial x^2}.$$

Apply $(\partial_t-\kappa\nabla^2)$ to both sides:

$$\left(\frac{\partial}{\partial t}-\nu\nabla^2\right)\left(\frac{\partial}{\partial t}-\kappa\nabla^2\right)\nabla^2 v_z'=g\alpha\frac{\partial^2}{\partial x^2}\left(\frac{\partial}{\partial t}-\kappa\nabla^2\right)T'.$$

Use (30) rearranged as $(\partial_t-\kappa\nabla^2)T'=(\Delta T/H)v_z'$:

$$\boxed{\left(\frac{\partial}{\partial t}-\nu\nabla^2\right)\left(\frac{\partial}{\partial t}-\kappa\nabla^2\right)\nabla^2 v_z'=\frac{g\alpha\Delta T}{H}\frac{\partial^2 v_z'}{\partial x^2}.} \tag{33}$$

(c) Normal modes and Rayleigh criterion

For $v_z'=A\sin(kx)\sin(\pi z/H)e^{\sigma t}$, the substitutions are $\nabla^2\to -K^2$ with $K^2=k^2+(\pi/H)^2$, $\partial_t\to\sigma$, $\partial_x^2\to -k^2$. Substitute into (33):

$$(\sigma+\nu K^2)(\sigma+\kappa K^2)(-K^2)\,v_z'=-\frac{g\alpha\Delta T}{H}\,k^2\,v_z'.$$

Cancel v_z' and the overall sign:

$$(\sigma+\nu K^2)(\sigma+\kappa K^2)K^2=\frac{g\alpha\Delta T}{H}k^2.$$

Divide by K^2:

$$(\sigma+\nu K^2)(\sigma+\kappa K^2)=\frac{g\alpha\Delta T}{H}\frac{k^2}{K^2}.$$

Expand the product on the left:

$$\sigma^2+(\nu+\kappa)K^2\sigma+\nu\kappa K^4=\frac{g\alpha\Delta T}{H}\frac{k^2}{K^2}.$$

Rearrange as a quadratic in σ:

$$\sigma^2+(\nu+\kappa)K^2\sigma+\nu\kappa K^4-\frac{g\alpha\Delta T}{H}\frac{k^2}{K^2}=0.$$

Apply the quadratic formula:

$$\boxed{\sigma = -\tfrac{1}{2}(\nu+\kappa)K^2 \pm \sqrt{\tfrac{1}{4}(\nu-\kappa)^2K^4 + \frac{g\alpha\Delta T}{H}\frac{k^2}{K^2}}.} \tag{34}$$

Instability $\sigma > 0$ requires the product of roots (constant term) to be negative:

$$\nu\kappa K^4 - \frac{g\alpha\Delta T}{H}\frac{k^2}{K^2} < 0.$$

Rearrange:

$$\frac{g\alpha\Delta T}{\nu\kappa H} > \frac{K^6}{k^2}. \tag{35}$$

Minimise $f(k) = (k^2+\pi^2/H^2)^3/k^2$ over k. Differentiate using the quotient rule:

$$\frac{\mathrm{d}f}{\mathrm{d}k} = \frac{3(k^2+\pi^2/H^2)^2\cdot 2k\cdot k^2 - (k^2+\pi^2/H^2)^3\cdot 2k}{k^4}.$$

Factor out $2k(k^2+\pi^2/H^2)^2/k^4$:

$$\frac{\mathrm{d}f}{\mathrm{d}k} = \frac{2(k^2+\pi^2/H^2)^2}{k^3}\left[3k^2 - (k^2+\pi^2/H^2)\right].$$

Set the bracket to zero:

$$3k^2 - k^2 - \pi^2/H^2 = 0.$$

Solve:

$$k_c^2 = \frac{\pi^2}{2H^2}.$$

Compute K_c^2:

$$K_c^2 = k_c^2 + \frac{\pi^2}{H^2} = \frac{\pi^2}{2H^2} + \frac{2\pi^2}{2H^2} = \frac{3\pi^2}{2H^2}.$$

Compute K_c^6:

$$K_c^6 = \left(\frac{3\pi^2}{2H^2}\right)^3 = \frac{27\pi^6}{8H^6}.$$

Form the threshold ratio:

$$\frac{K_c^6}{k_c^2} = \frac{27\pi^6/(8H^6)}{\pi^2/(2H^2)} = \frac{27\pi^6}{8H^6}\cdot\frac{2H^2}{\pi^2} = \frac{27\pi^4}{4H^4}.$$

Substitute into the instability condition $g\alpha\Delta T/(\nu\kappa H) > K_c^6/k_c^2$ and multiply by H^4:

$$\boxed{\mathrm{Ra} \equiv \frac{g\alpha\Delta T\, H^3}{\nu\kappa} > \frac{27\pi^4}{4} \approx 657.5.} \tag{36}$$

This applies to stress-free isothermal boundaries (the sine ansatz). Rigid no-slip raises $\mathrm{Ra}_c \approx 1708$. Critical wavelength $\lambda_c = 2\pi/k_c = 2\sqrt{2}\,H$: rolls of horizontal extent $\sim H$. At threshold $\sigma = 0$ (steady bifurcation).